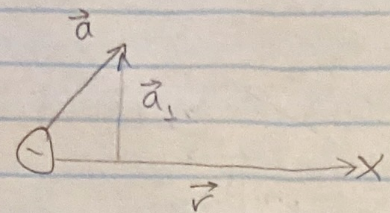
WPS 22 CRSALEO10

1. 

Now $\hat{E}_{rad} = -q\hat{a}_\perp$ so $\hat{E}_{rad} = \hat{a}_\perp$ so (a)

2. We have that $B = \frac{E}{c}$ with $\hat{E} \perp \hat{R}$.
so $B = \frac{10^4}{3\times10^8} \Rightarrow B = 3.33\times10^{-5}\ T$
Since $\hat{v} = \hat{k}$ and $\hat{E} = \hat{\imath}$, we have that $\hat{B} = \hat{\jmath}$ so
$\vec{B} = \langle 0, 3\times10^4, 0\rangle T$ (check)

3a) $I_D = \varepsilon_0 \cdot \frac{d\Phi_{elec}}{dt} \cdot \mu_0$
$= \varepsilon_0 \frac{d}{dt} \int \vec{E}\cdot\hat{n}\, dA \cdot \mu_0$
$= \varepsilon_0 \cdot \frac{d}{dt} \left[(10^5 - 10^4 t)(\pi (0.1)^2)\right] \cdot \mu_0$
$= -\varepsilon_0 \cdot \pi \cdot 0.1^2 \cdot 10^4 \cdot \mu_0 = -3.48\times10^{-15}$

b) The negative indicates I_D is down

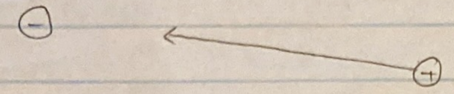
c) Using the right hand rule, the magnetic field is clockwise as viewed from above

4a) Using the circular path:
$\oint \vec{B}\cdot d\vec{l} = \mu_0\left[\Sigma I_{inside} + \varepsilon_0 \frac{d}{dt}\int \vec{E}\cdot\hat{n}\, dA\right]$
$\Leftrightarrow B(2\pi R) = \mu_0 \left[I_{in} + \varepsilon_0 \cdot 0\right]$
$= \mu_0 N I$ (for N turns)
So $B = \frac{\mu_0 I N}{2\pi R}$

b) $\oint \vec{B}\cdot d\vec{l} = \mu_0\left[\Sigma I_{inside} + \varepsilon_0 \cdot \frac{d}{dt}\int \vec{E}\cdot\hat{n}\, dA\right]$
$B(2\pi R) = \mu_0 N I + \varepsilon_0 \cdot \frac{d}{dt}\left(E\cdot \pi R^2\right)]$
$= \mu_0 N I + \varepsilon_0 \cdot \frac{dE}{dt}\cdot \pi R^2$

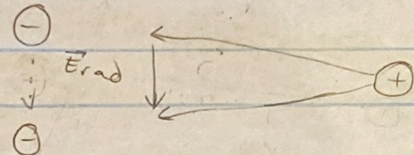
So $B = \dfrac{\mu_0 N I + \varepsilon_0 \cdot \frac{dE}{dt} \cdot \pi R^2}{2\pi R}$

5. Initially the proton experiences an electric field in the direction:



But this is negligible (very large distance)

As the electron is accelerated downwards:



So the proton experiences a radiative electric fields and thus a force downwards over time Δt. Now the proton is moving so the proton also experiences a radiative magnetic field to the right since $\vec{E}_{rad}$ propagates to the right, with $\hat{\jmath}$ down so $\hat{U} = \vec{E} \times \vec{B}$ direction so $\hat{B} = \hat{k}$ and thus $\vec{F} = q\vec{v} \times \vec{B}$ is to the right. After Δt, only the coulombic force acts on the proton which is negligible so the proton moves at a constant speed.

6a) We have that $\ell = c\Delta t$

$= 3 \times 10^8 \times 20 \times 10^{-9}$

$= 6\ \text{m}$

b) Energy density $= \dfrac{E}{V}$

$= \dfrac{10}{\pi(0.001^2)(6)}$

$= 5.31 \times 10^5\ \text{J/m}^3$

c) We know that Energy density $= \varepsilon_0 E^2$

So $5.31\times10^5 = \varepsilon_0 E^2 = 8.85\times10^{-12} E^2 \Rightarrow E = 2.45\times10^8 \text{ N/C}$

Thus $B = \frac{E}{c} = \frac{3\times10^8}{2.45\times10^8} = \cancel{1.23}\ 0.813\text{T}$

7a) The solar sail depends on the electromagnetic momentum from the radiation of the sun and thus a larger force needs a larger intensity. A reflecting surface means double the pressure so closer means greater intensity. Both gravitational and electromagnetic force thus depend inversely on r^2, but with a small mass and large area, closer is more efficient.

b) For an ideally reflecting surface, we have

$P = 2\cdot\frac{\vec{S}}{c}$, so $\vec{F} = 2\cdot\frac{\vec{S}}{c}A$

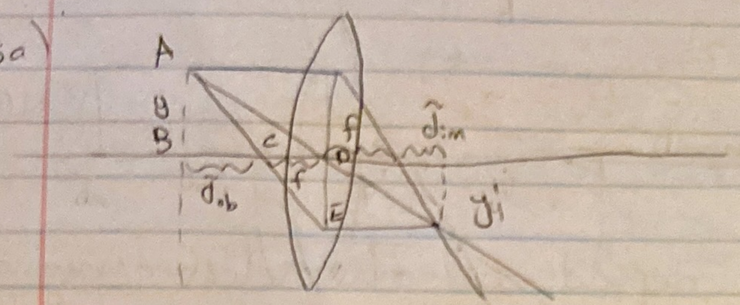
Thus we need $2\cdot\frac{\vec{S}}{c}A = G\frac{M_1 M_2}{R^2}$

So $2\cdot\frac{I}{c}A = G\frac{M_1 M_2}{R^2}$ where I is the intensity

$\Leftrightarrow 2\cdot\frac{1400}{3\times10^8}A = 6.7\times10^{-11}\cdot\frac{103\cdot2\cdot10^{30}}{(1.5\times10^{11})^2}$

$\Leftrightarrow A = 6.38\times10^5 \text{ m}^2$

8a)



b) $\tilde{d}_{ob} \cdot \tilde{d}_{im} = (d_{ob} - f)(d_{im} - f)$

$= d_{ob} d_{im} - f(d_{ob} + d_{im}) + f^2$

Now $d_{ob} d_{im} - f(d_{ob} + d_{im}) = d_{ob} d_{im} - \frac{1}{\frac{1}{d_{ob}} + \frac{1}{d_{im}}}(d_{ob} + d_{im})$

$= d_{ob} d_{im} - \frac{d_{ob} d_{im}}{d_{ob} + d_{im}}(d_{ob} + d_{im})$

$= d_{ob} d_{im} - d_{ob} d_{im} = 0$

So $\tilde{d}_{ob} \tilde{d}_{im} = f^2$

c) $m = -\frac{y'}{y}$

Now $\Delta ABC \,/\!/\!/\, \Delta EDC$

So $\frac{AB}{ED} = \frac{BC}{DC}$

$\Leftrightarrow \frac{y}{y'} = \frac{\tilde{d}_{ob}}{f}$

$\Leftrightarrow \frac{y'}{y} = \frac{f}{\tilde{d}_{ob}}$

So $m = -\frac{y'}{y} = \frac{-f}{\tilde{d}_{ob}} = -\frac{f \tilde{d}_{im}}{\tilde{d}_{ob} \tilde{d}_{im}}$

$= -\frac{f \tilde{d}_{im}}{f^2}$

$= -\frac{\tilde{d}_{im}}{f}$